

AULYN -- Data Structures & Algorithms: Binary Search Trees

Course: CS201 | Instructor: Prof. Sarah Jenkins

1. Binary Search Tree (BST) Definition:

- A binary tree where for every node X:
 - * Left subtree contains values strictly smaller than X
 - * Right subtree contains values strictly greater than X

2. Time Complexities:

- Search / Insert / Delete (Balanced BST): $O(\log N)$
- Search / Insert / Delete (Skewed Tree): $O(N)$

3. In-order Traversal Property:

- An In-order traversal (Left $\rightarrow$ Node $\rightarrow$ Right) of a BST visits nodes in strictly sorted order.